

- 8 (a) neutron is a single nucleon / particle [1]
- (b) binding energy = $4 \times 7.07 \times 1.6 \times 10^{-13}$ C1
 $= 4.52 \times 10^{-12} \text{ J}$
 binding energy = $c^2 \Delta m$ C1
 $4.52 \times 10^{-12} = (3.0 \times 10^8)^2 \times \Delta m$
 $\Delta m = 5.03 \times 10^{-29} \text{ kg}$ A1 [3]
- (c) (i) fusion(do not allow fussion) B1 [1]
- (ii) $(2 \times 1.12) + 3x = 28.28$ C1
 $\dots -17.7$ C1
 $x = 2.78 \text{ MeV per nucleon}$ A1 [3]
 (use of +17.7 gives $x = 14.6 \text{ MeV}$, allow 1 mark only)

[Total: 8]

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Section B

9. (a) resistance of wire = l/A

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